

HARMANPREET KAUR

Ai & Machine Learning Undergraduate | Frontend developer intern

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EDUCATION

Chandigarh University | Mohali, Punjab
B.E. in Computer Science and Engineering (AI&ML)

July 2024 – July 2028
CGPA: 8.35/10

EXPERIENCE

Frontend Developer Intern / *Trigunasoftware (Remote)*

July 2025 - Present

Tech: Angular, TypeScript, SignalR, JWT Auth, Angular Signals

- Developed 31+ components and 31 routed screens across user, content, and admin modules.
- Implemented JWT authentication and route guards, securing 90%+ of application routes.
- Built multilingual support (English/German), real-time chat with SignalR, and admin dashboards for content management.
- Leveraged Angular Signals across 100+ state points to enhance responsiveness and state management.

PROJECTS

LOST & FOUND AI PLATFORM

[Github](#)

Tech: Next.js, FastAPI, MongoDB, WebSockets, Gemini API, Hugging Face

- Built an AI-powered platform for matching lost items with finders using semantic, location, and image-based analysis.
- Reduced server memory usage by over 90% by offloading AI inference to Hugging Face and Google Gemini APIs and implementing lazy model loading.
- Improved API performance by 40% through asynchronous FastAPI architecture and MongoDB Motor integration.
- Reduced messaging bandwidth overhead by 85% using WebSockets, enabling real-time communication with sub-50ms message delivery.
- Designed a secure claim verification workflow that reduced unsolicited messages and spam by approximately 95%.

CINEMIND – AI-Powered Movie Recommendation System

[Github](#)

Tech: Python, Pandas, Scikit-Learn, TF-IDF, Cosine Similarity

- Developed a content-based recommendation engine using TF-IDF vectorization and cosine similarity.
- Performed data preprocessing and feature engineering on movie metadata.
- Generated personalized movie recommendations through similarity-based retrieval techniques.

Image Caption Generator

[Github](#)

Tech: Python, TensorFlow, CNN, LSTM

- Developed an image captioning system using CNN-LSTM to generate descriptive captions from images.
- Trained on the Flickr8K dataset using transfer learning, feature extraction, and sequence modeling.
- Built a web interface for real-time image upload and caption generation.

TECHNICAL SKILLS

- Languages:** Python, C++, JavaScript, TypeScript, SQL
- Frontend:** React, Next.js, Angular, HTML, CSS, Tailwind CSS, Bootstrap
- Backend:** FastAPI, REST APIs, JWT Authentication, OAuth 2.0, WebSockets
- AI/ML:** Data Preprocessing, EDA, Feature Engineering, Machine Learning, Deep Learning, NLP, Recommendation Systems, Model Evaluation
- Databases:** MongoDB, Microsoft SQL Server
- Tools:** Git, GitHub, Postman, VS Code

ACHIEVEMENTS

- Awarded a 100% scholarship for the entire four-year undergraduate program based on academic merit and potential.
- Participated in college-level hackathons and collaborative technical competitions.
- Won multiple school-level competitions in academics, singing, painting, and sports.
- Explored research in Machine Learning, Deep Learning, NLP, and AI applications.